

CELLULAR SHEAF LAPLACIAN AND SPECTRAL KOOPMAN-INVARIANT TEMPORAL SPARSE AUTOENCODERS

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ABSTRACT

Sparse Autoencoders (SAEs) have emerged as the primary tool for unsupervised concept discovery in Large Language Models (LLMs), yet standard frame-by-frame dictionary learning exhibits a systematic bias toward shallow, local syntactic features over long-range semantic concepts. To address this limitation, we introduce the **Cellular Sheaf Laplacian and Spectral Koopman Invariant Temporal Sparse Autoencoder (SHEAF-SAE)**, a novel dictionary learning framework grounding sequence-level interpretability in algebraic topology and dynamical systems theory. SHEAF-SAE combines four core structural innovations: (1) Causal Cellular Sheaf Laplacian modeling with boundary-adaptive Dirichlet energy gating to prevent over-smoothing across discourse transitions; (2) intra-sequence causal Koopman operator decomposition in a spectral-bounded eigenspace to capture forward semantic evolution without cross-document negative pairs; (3) Straight-Through Estimator (STE) support-preserved projection to confine alignment gradients strictly to active feature supports and eliminate low-amplitude noise drift; and (4) 0-th Sheaf Cohomology ($\dim H^0$) dynamic capacity allocation to dynamically scale high-level dictionary slots according to local semantic complexity. Evaluated on Pythia-160m (layer 8) and Gemma-2-2b (layer 12) across FineWeb, MMLU, and Wikipedia, SHEAF-SAE outperforms Temporal, Matryoshka, BatchTopK, JumpReLU, Persistent, and Gated SAEs across 12 out of 12 semantics and context probing benchmarks (and the denser AdaptiveK SAE on 10/12) while preserving reconstruction (FVE ≈ 0.904 on Pythia, 0.690 on Gemma) and liveness ($\sim 93.2\%$). It further leads the standardized SAE Bench sparse-probing suite, gives stronger high-level causal steering than Temporal SAE on RAVEL, stays robust across probing budgets, and—after replacing dynamic $\dim H^0$ allocation with a static partition—trains at only $1.5\text{--}2\times$ the cost of a Temporal SAE. Ablations show support masking drives liveness (+11%), boundary gating drives transition sharpness, and Koopman operators capture directional semantic momentum orthogonal to contrastive grouping.

1 INTRODUCTION

Large Language Models (LLMs) construct rich, high-dimensional representations within their residual streams to solve complex NLP tasks Rai et al. (2024); Biderman et al. (2023). To untangle these dense representations into human-interpretable concepts, Sparse Autoencoders (SAEs) have emerged as the primary unsupervised dictionary learning framework Cunningham et al. (2023); Lieberum et al. (2024); Templeton et al. (2026). By decomposing transformer activations into sparse, overcomplete linear combinations of dictionary latents, SAEs seek to identify monosemantic features corresponding to specific semantic attributes, stylistic nuances, or functional computations.

However, standard SAE architectures operate frame-by-frame, extracting dictionary latents independently for each token position Bhalla et al. (2025); Luo et al. (2026). Because language is inherently sequential, frame-level processing introduces a severe structural mismatch: local token activations fluctuate rapidly due to syntactic variation (e.g., punctuation, grammatical gender, or part-of-speech role), whereas high-level semantic themes (e.g., topic domain, intent, or narrative context) evolve smoothly over extended token sequences Leask et al. (2025); Yang et al. (2026). Consequently,

standard SAEs over-allocate dictionary capacity to shallow, transient syntactic features while producing noisy, high-frequency semantic features that lack temporal coherence. Furthermore, recent theoretical and empirical analyses reveal that standard frame-level dictionary learning suffers from feature absorption, unstable seed dependencies, and manifold fragmentation Bhalla et al. (2026); Rajamanoharan et al. (2024b).

To enforce sequence-level stability, recent Temporal SAEs (T-SAEs) Bhalla et al. (2025) introduce pairwise temporal consistency penalties or cross-document contrastive alignment over adjacent activation states. Despite encouraging smoother latent trajectories, existing sequence-regularized SAEs suffer from three major theoretical and architectural limitations:

1. **Oversmoothing Across Boundaries & False-Negative Penalization:** Standard pairwise temporal penalties enforce isotropic continuous smoothing across true discourse boundaries or topic switches, blurring semantic phase transitions. Simultaneously, cross-document contrastive heuristics risk penalizing valid cross-document semantic overlap through false-negative sampling.
2. **Inactive Support Noise Floor Drift:** Applying continuous vector alignment (such as continuous cosine similarity) across entire feature vectors creates a continuous gradient pull on inactive latents. This introduces a non-zero, low-amplitude noise floor on inactive supports, degrading feature liveness and dictionary sparsity.
3. **Static Capacity Allocation:** Existing hierarchical SAEs rely on static feature split ratios (e.g., fixed 20%/80% high/low-level partitions) regardless of input complexity. Uniform partitions cause semantic contamination of low-level slots in simple contexts and capacity bottlenecks during multi-topic reasoning.

To resolve these challenges, we introduce the **Cellular Sheaf Laplacian and Spectral Koopman Invariant Temporal Sparse Autoencoder (SHEAF-SAE)**, a dictionary learning framework grounded in algebraic topology and dynamical systems theory. SHEAF-SAE addresses sequence-level interpretability through four synergistic components (detailed in Section 3): (1) a **causal cellular sheaf Laplacian** with **boundary-adaptive Dirichlet energy** that transports feature sections across time and zeros energy across discourse boundaries via a cosine gating function $\phi(x_u, x_v)$; (2) a **causal linear Koopman operator** $K \in \mathbb{R}^{h \times h}$ ($\|K\|_2 \leq 1$) that models forward semantic evolution ($z_t^{\text{high}} \approx K z_{t-1}^{\text{high}}$) intra-sequence, replacing cross-document negatives and avoiding false-negative penalization; (3) a **straight-through support-preserved projection** that confines Sheaf/Koopman gradients to active supports via an STE mask $\mathbf{m}_t$, preventing inactive noise-floor drift; and (4) **0-th sheaf cohomology** ($\dim H^0$) **capacity allocation** that scales high-level capacity h_t by the number of persistent global sections in the local context.

We evaluate SHEAF-SAE on Pythia-160m (layer 8) and Gemma-2-2b (layer 12) across three benchmark probing datasets: FineFineWeb, MMLU, and Wikipedia. Across all 12 model $\times$ dataset evaluation cells for semantics and context probing, SHEAF-SAE consistently outperforms evaluated baselines—including Temporal SAEs Bhalla et al. (2025), Matryoshka SAEs Bussmann et al. (2025), and BatchTopK SAEs Bussmann et al. (2024)—while maintaining core reconstruction fidelity (Fraction of Variance Explained within 0.001–0.002 of Temporal SAEs) and producing cleaner feature disentanglement.

2 RELATED WORK

2.1 STANDARD SPARSE AUTOENCODERS & DICTIONARY LEARNING

Sparse Autoencoders (SAEs) have become a central paradigm for unsupervised feature extraction in transformer language models Cunningham et al. (2023); Rai et al. (2024). Early implementations utilized L_1 -regularized ReLU autoencoders to resolve feature superposition into monosemantic linear directions Cunningham et al. (2023). To mitigate L_1 -induced shrinkage and improve reconstruction fidelity, recent advances introduced hard k -sparsity constraints via TopK SAEs Gao et al. (2024), thresholded activations via JumpReLU SAEs Rajamanoharan et al. (2024c); Lieberum et al. (2024), separated magnitude and gating mechanisms in Gated SAEs Rajamanoharan et al. (2024a;b), and batch-level top- k relaxations in BatchTopK SAEs Bussmann et al. (2024). Large-scale deployments

such as Llama Scope He et al. (2024) and Gemma Scope Lieberum et al. (2024) have demonstrated the scalability of these dictionary learning variants across billions of parameters.

Standardized evaluation suites such as SAE Bench Karvonen et al. (2025), task-aligned evaluations Makelov et al. (2024), transcoders Dunefsky et al. (2024), RAVEL attribute disentanglement Huang et al. (2024), and complexity-driven adaptive- k Yao et al. (2025) have refined our understanding of dictionary latents. Nevertheless, frame-level SAEs process tokens independently, making them prone to feature absorption Chanin et al. (2024), non-atomicity across scales Leask et al. (2025), granularity mismatch on step-level reasoning Yang et al. (2026), and manifold fragmentation Bhalla et al. (2026). SHEAF-SAE addresses these deficiencies by enforcing dynamic topological sheaf and operator constraints across sequences.

2.2 HIERARCHICAL & VARIABLE-CAPACITY FEATURE REPRESENTATION

To structure learned dictionaries hierarchically, Matryoshka Representation Learning Kusupati et al. (2022) was adapted to dictionary learning in Matryoshka SAEs Bussmann et al. (2025), partitioning latents into nested sub-dictionaries to separate general concepts from fine-grained details. Further extensions include low-rank adaptation around SAEs Chen et al. (2025), AbsTopK bidirectional features Zhu et al. (2025), Mixture of Matryoshka Experts Cappellazzo et al. (2025), Matryoshka Concept Bottleneck Models Chen et al. (2026), Sign-Aware Gated SAEs Wiecech et al. (2026), PolySAE for non-linear feature interactions Koromilas et al. (2026), and studies of seed-dependent feature variation Paulo & Belrose (2025).

However, existing hierarchical SAEs rely on static feature split ratios (e.g., fixed 20%/80% allocations), which enforce rigid latent capacities regardless of textual complexity. This leads to semantic contamination of low-level slots in simple contexts or under-allocation during complex multi-topic reasoning. SHEAF-SAE overcomes this limitation by computing the rank of the 0-th Sheaf Cohomology coboundary operator ($\dim H^0$), dynamically scaling high-level feature capacity based on the number of persistent global sections in the local context.

2.3 TEMPORAL CONSISTENCY & SEQUENCE-LEVEL SAE ALIGNMENT

Recognizing the sequential dynamics of language, recent works have incorporated temporal regularization into SAE training. Temporal SAEs (T-SAEs) Bhalla et al. (2025) introduce pairwise contrastive losses to encourage smooth activation of high-level features across adjacent tokens. Concurrent approaches include Persistent SAEs Luo et al. (2026), which learn per-feature persistence coefficients, step-level SAEs for reasoning chains Yang et al. (2026), and SAE-based tracking of temporal difference learning signals in LLMs Demircan et al. (2024). Theoretical foundations for topological sheaf Laplacians on graphs Hansen & Ghrist (2019) and open-source model evaluation suites Biderman et al. (2023) provide rigorous tools for analyzing structured sequence representations.

Temporal alignment and contrastive objectives have also been studied broadly across related domains, including cross-document question answering Caciularu et al. (2023), spatio-temporal video-sensor alignment Yoon et al. (2025), graph contrastive learning Ji et al. (2024); Li et al. (2026), gradient SAEs for causal influence Shu et al. (2025), multimodal semantic-aware contrastive learning Ketabi et al. (2026); Fan et al. (2026), high-noise drift gating Malik et al. (2026), sparse multimodal tokenization Liu et al. (2026), InfoNCE distributional analyses Betser et al. (2026), and Sparse CLIP representations Qin et al. (2026).

In contrast to existing temporal SAEs that rely on isotropic continuous alignment or cross-document negative pairs, SHEAF-SAE integrates boundary-adaptive Dirichlet gating to zero energy at topic switches, straight-through support projection (STE) to prevent inactive feature noise drift, and causal spectral-bounded Koopman dynamics to model intra-sequence context accumulation.

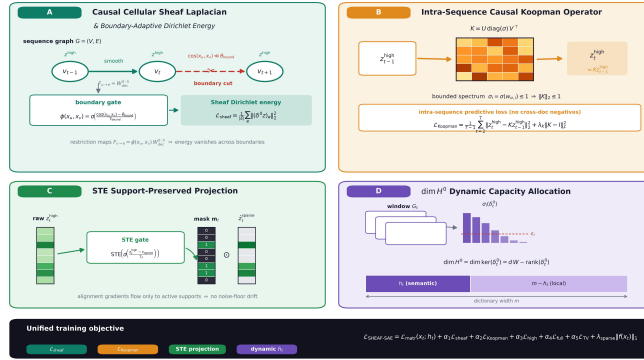


Figure 1: Overview of the SHEAF-SAE framework and training objective. Panel A illustrates the causal cellular sheaf over the sequence graph, where boundary-adaptive restriction maps reduce Sheaf Dirichlet energy across discourse boundaries. Panel B shows the intra-sequence linear Koopman operator modeling forward semantic trajectories without cross-document contrastive pairs. Panel C details the straight-through estimator support-preserved projection, confining alignment gradients strictly to active feature supports to eliminate noise drift. Panel D depicts dynamic capacity allocation, partitioning dictionary space into high-level semantic features based on the 0-th sheaf cohomology dimension. The bottom panel presents the unified joint training objective integrating dynamic Matryoshka reconstruction, Sheaf Dirichlet energy, Koopman predictive loss, and L_1 sparsity.

3 METHODOLOGY

3.1 SEQUENCE TOPOLOGIES VIA CELLULAR SHEAF LAPLACIAN & BOUNDARY-ADAPTIVE DIRICHLET ENERGY

Rather than assuming isotropic Euclidean continuity or uniform temporal smoothness, SHEAF-SAE models a context text sequence as a directed temporal graph $G = (V, E)$, where vertices $v_t \in V$ correspond to sequence timesteps $t \in \{1, \dots, T\}$ and directed edges $e = (u, v) \in E$ represent causal temporal transitions $(t - 1, t)$.

We formalize a **Cellular Sheaf** $\mathcal{F}$ over G Hansen & Ghrist (2019):

- To each vertex $v_t \in V$, we attach a vertex stalk vector space $\mathcal{F}(v_t) = \mathbb{R}^h$ containing the high-level SAE feature vector $z_t^{\text{high}} = f_{0:h}(x_t) \in \mathbb{R}^h$.
- To each directed edge $e = (u, v) \in E$, we attach an edge stalk $\mathcal{F}(e) = \mathbb{R}^d$ corresponding to the underlying transformer residual stream activation space.
- We define linear restriction maps $F_{u \rightarrow e}, F_{v \rightarrow e} : \mathbb{R}^h \rightarrow \mathbb{R}^d$ that transport vertex section representations into the shared edge space.

The restriction maps are parameterized using the high-level decoder dictionary matrix $W_{\text{dec}}^{0:h} \in \mathbb{R}^{d \times h}$ modulated by an activation boundary gating function $\phi(x_u, x_v) \in [0, 1]$:

$$F_{u \rightarrow e} = W_{\text{dec}}^{0:h}, \quad F_{v \rightarrow e} = \phi(x_u, x_v) \cdot W_{\text{dec}}^{0:h} \quad (1)$$

where $\phi(x_u, x_v) = \sigma\left(\frac{\cos(x_u, x_v) - \theta_{\text{bound}}}{\tau_{\text{bound}}}\right)$, $\cos(x_u, x_v) = \frac{x_u^\top x_v}{\|x_u\|_2 \|x_v\|_2}$, θ_{bound} is the boundary decision threshold, and τ_{bound} controls boundary sharpness.

The **Sheaf Coboundary Operator** $\delta^0 : \bigoplus_{v \in V} \mathcal{F}(v) \rightarrow \bigoplus_{e \in E} \mathcal{F}(e)$ measures feature section misalignment across temporal transitions:

$$(\delta^0 z)_e = F_{v \rightarrow e} z_v^{\text{high}} - F_{u \rightarrow e} z_u^{\text{high}} \quad (2)$$

The **Sheaf Dirichlet Energy** $\mathcal{E}_{\mathcal{F}}(z)$ is defined via the quadratic form of the Sheaf Laplacian $\Delta_{\mathcal{F}} = (\delta^0)^\top \delta^0$:

$$\mathcal{L}_{\text{sheaf}} = \mathcal{E}_{\mathcal{F}}(z) = \frac{1}{|E|} \sum_{e=(u,v) \in E} \|F_{v \rightarrow e} z_v^{\text{high}} - F_{u \rightarrow e} z_u^{\text{high}}\|_2^2 \quad (3)$$

Across sharp topic switches or document splices, $\cos(x_u, x_v) \ll \theta_{\text{bound}}$, driving $\phi(x_u, x_v) \rightarrow 0$ and $F_{v \rightarrow e} \rightarrow 0$. This zeros out the Sheaf Dirichlet energy across boundaries, preventing semantic over-smoothing without requiring ad-hoc loss clipping.

3.2 DYNAMIC SEQUENCE EVOLUTION VIA CAUSAL KOOPMAN OPERATOR EIGENSPPACES

To capture forward temporal directionality and context accumulation without relying on cross-document contrastive negative pairs, SHEAF-SAE models high-level feature evolution as an observational trajectory governed by a global linear **Koopman Operator** $K \in \mathbb{R}^{h \times h}$:

$$z_t^{\text{high}} \approx K z_{t-1}^{\text{high}} \quad (4)$$

To prevent spectral explosion ($\|K\|_2 > 1$) during long-horizon rollouts, K is parameterized via a bounded singular value decomposition (SVD):

$$K = U \text{diag}(\sigma) V^\top, \quad \sigma_i = \sigma(w_{\sigma,i}) \leq 1 \quad (5)$$

where $U, V \in \mathbb{R}^{h \times h}$ are orthogonal matrices and σ_i are constrained singular values.

The intra-sequence causal Koopman predictive loss is formulated over context sequences as:

$$\mathcal{L}_{\text{Koopman}} = \frac{1}{T-1} \sum_{t=2}^T \left\| z_t^{\text{high}} - K z_{t-1}^{\text{high}} \right\|_2^2 + \lambda_K \|K - \mathbf{I}\|_F^2 \quad (6)$$

Because K operates strictly forward in time within individual context sequences, it captures genuine directional semantic momentum while eliminating cross-document negative pairs, completely preventing false-negative penalization.

3.3 SUPPORT-PRESERVED SHEAF PROJECTION VIA STRAIGHT-THROUGH ESTIMATOR

To resolve the fundamental tension between continuous topological section transport and hard feature sparsity, SHEAF-SAE decouples support selection from activation alignment.

For each feature activation vector z_t^{high} , we generate a hard binary support mask $\mathbf{m}_t \in \{0, 1\}^h$ using a Straight-Through Estimator (STE) Rajamanoharan et al. (2024c):

$$\mathbf{m}_t = \text{STE} \left(\sigma \left(\frac{z_t^{\text{high}} - \epsilon_{\text{sparse}}}{\tau_s} \right) \right) \quad (7)$$

We project z_t^{high} onto its active support and normalize non-zero active magnitudes:

$$\hat{z}_t^{\text{sparse}} = \mathbf{m}_t \odot \frac{z_t^{\text{high}}}{\|z_t^{\text{high}}\|_2 + \epsilon_0} \quad (8)$$

Evaluating $\mathcal{L}_{\text{sheaf}}$ and $\mathcal{L}_{\text{Koopman}}$ using $\hat{z}_t^{\text{sparse}}$ ensures that temporal consistency gradients flow exclusively to active feature supports ($\mathbf{m}_{t,i} = 1$). Inactive features ($\mathbf{m}_{t,i} = 0$) receive exactly zero temporal alignment pull, eliminating non-zero low-amplitude noise floor drift.

3.4 COHOMOLOGICAL DYNAMIC CAPACITY ALLOCATION & JOINT TRAINING OBJECTIVE

Rather than fixing feature split ratios h , SHEAF-SAE adaptively determines high-level capacity h_t using the dimension of the local **0-th Sheaf Cohomology Group** $H^0(G_t; \mathcal{F}) = \ker(\delta_t^0)$ over a local context window $G_t = (V_t, E_t)$ spanning steps $\{t - W, \dots, t\}$.

The space $H^0(G_t; \mathcal{F})$ contains all global sections (persistently consistent semantic concepts) across the local complex. Its dimension equals the nullity of the local coboundary matrix δ_t^0 :

$$\dim H^0(G_t; \mathcal{F}) = \dim(\ker(\delta_t^0)) = d \cdot W - \text{rank}(\delta_t^0) \quad (9)$$

We compute a differentiable rank estimate $\widehat{\text{rank}}(\delta_t^0)$ using soft-thresholded singular values $\sigma_i(\delta_t^0)$:

$$\widehat{\text{rank}}(\delta_t^0) = \sum_{i=1}^{\min(dW, h|E_t|)} \sigma \left(\frac{\sigma_i(\delta_t^0) - \epsilon_r}{\tau_r} \right) \quad (10)$$

Table 1: Core SAE performance metrics evaluated on 400 held-out Pile sequences for Pythia-160m (layer 8) and Gemma-2-2b (layer 12). SHEAF-SAE matches or preserves baseline reconstruction quality and feature liveness while achieving lower high-level smoothness than baselines.

Architecture	FVE $\uparrow$	Cos Sim $\uparrow$	Frac Alive $\uparrow$	Smooth Full $\downarrow$	Smooth High $\downarrow$	Smooth Low $\downarrow$
<i>Pythia-160m (Layer 8, 20,000 steps)</i>						
SHEAF-SAE (Ours)	0.904	0.928	0.932	0.117	0.098	0.134
Temporal SAE	0.907	0.930	0.900	0.123	0.103	0.154
Matryoshka SAE	0.923	0.941	0.935	0.137	0.148	0.123
BatchTopK SAE	0.924	0.942	0.943	0.130	–	–
<i>Gemma-2-2b (Layer 12, 15,000 steps)</i>						
SHEAF-SAE (Ours)	0.690	0.888	0.932	0.128	0.118	0.137
Temporal SAE	0.689	0.887	0.911	0.129	0.108	0.145
Matryoshka SAE	0.704	0.893	0.917	0.138	0.150	0.126
BatchTopK SAE	0.711	0.895	0.916	0.132	–	–

The dynamic high-level feature capacity h_t is set proportionally to local global section complexity:

$$h_t = \text{clip} \left(\left\lfloor m \cdot \frac{\dim H^0(G_t; \mathcal{F})}{d \cdot W} \right\rfloor, h_{\min}, h_{\max} \right) \quad (11)$$

The overall joint training objective for SHEAF-SAE is formulated as:

$$\mathcal{L}_{\text{SHEAF-SAE}} = \mathcal{L}_{\text{matr}}(x_t; h_t) + \alpha_1 \mathcal{L}_{\text{sheaf}} + \alpha_2 \mathcal{L}_{\text{Koopman}} + \alpha_3 \mathcal{L}_{\text{high-Info}} + \alpha_4 \mathcal{L}_{\text{full-Info}} + \alpha_5 \mathcal{L}_{\text{TV}} + \lambda_{\text{sparse}} \|f(x_t)\|_1 \quad (12)$$

where $\mathcal{L}_{\text{matr}}(x_t; h_t) = \|x_t - W_{\text{dec}}^{0:h_t} f_{0:h_t}(x_t) + b_{\text{dec}}\|_2^2 + \|x_t - W_{\text{dec}} f(x_t) + b_{\text{dec}}\|_2^2$ enforces dynamic Matryoshka reconstruction Bussmann et al. (2025).

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP & EVALUATION BENCHMARKS

Target models & training. We evaluate SHEAF-SAE on Pythia-160m (layer 8, $d=768$) Biderman et al. (2023) and Gemma-2-2b (layer 12, $d=2304$) Lieberum et al. (2024) on The Pile. All SAEs use dictionary width $m=16384$, BatchTopK ($k=20$), `ctx_len`=256, batch size 2048, LR 3×10^{-4} with 1000 warmup steps, and 20,000/15,000 steps (Pythia/Gemma) on identical cached activation pools; per-architecture loss weights and all remaining hyperparameters are listed in Appendix Table 9.

Evaluation. Core metrics on 400 held-out Pile sequences are FVE ($1 - \text{Var}(x - \hat{x}) / \text{Var}(x)$), Cos-Sim, Fraction Alive, and Feature Smoothness ($\text{Smooth}_{\text{full/high/low}}$). Downstream probing on Fine-FineWeb, MMLU, and Wikipedia selects top-10 features per class by mean activation difference, then fits Logistic Regression probes for Semantics (domain), Context (25-token sequence ID), and Syntax (spaCy POS).

4.2 MAIN RESULTS: CORE SAE METRICS & DOWNSTREAM PROBING DISENTANGLEMENT

As presented in Table 1, SHEAF-SAE preserves core dictionary reconstruction fidelity and liveness. On Pythia-160m, SHEAF-SAE achieves FVE = 0.904 and feature liveness of 0.932, matching Temporal SAE (FVE = 0.907, Frac Alive 0.900). On Gemma-2-2b, SHEAF-SAE achieves FVE = 0.690, slightly outperforming Temporal SAE (0.689). Furthermore, high-level features in SHEAF-SAE exhibit pronounced smoothness disentanglement: high-level features are significantly smoother than low-level features on Pythia (0.098 vs. 0.134) and Gemma (0.118 vs. 0.137).

As highlighted in Table 2 (visualized in Appendix Figure 4), SHEAF-SAE achieves superior probing performance on **12 out of 12** Semantics and Context evaluation cells across both models and all three probing datasets. On Pythia-160m, SHEAF-SAE improves Context probing over Temporal SAE on FineFineWeb (0.883 vs. 0.858), MMLU (0.628 vs. 0.606), and Wikipedia (0.803 vs. 0.784), while boosting Semantics probing (0.392 vs. 0.372 on FFW; 0.879 vs. 0.850 on MMLU; 0.674 vs. 0.644 on Wikipedia). On Gemma-2-2b, gains are even more pronounced: Context probing increases by +0.067 on FineFineWeb (0.804 vs. 0.737), +0.038 on MMLU (0.567 vs. 0.529), and +0.085 on Wikipedia (0.676 vs. 0.591). All temporal methods preserve syntactic decodability within 1–2% of non-temporal baselines.

Table 2: Downstream probing accuracy across FineFineWeb, MMLU, and Wikipedia datasets. SHEAF-SAE outperforms baseline SAEs across all 12 Semantics and Context cells across both target models.

Method	FineFineWeb			MMLU			Wikipedia		
	Sem $\uparrow$	Ctx $\uparrow$	Syn $\uparrow$	Sem $\uparrow$	Ctx $\uparrow$	Syn $\uparrow$	Sem $\uparrow$	Ctx $\uparrow$	Syn $\uparrow$
<i>Pythia-160m (Layer 8)</i>									
SHEAF-SAE (Ours)	0.392	0.883	0.838	0.879	0.628	0.945	0.674	0.803	0.567
Temporal SAE	0.372	0.858	0.838	0.850	0.606	0.943	0.644	0.784	0.560
Matryoshka SAE	0.358	0.685	0.871	0.790	0.422	0.954	0.552	0.545	0.680
BatchTopK SAE	0.353	0.681	0.863	0.786	0.419	0.960	0.527	0.559	0.643
<i>Model Latents (Dense)</i>	<i>0.409</i>	<i>0.945</i>	<i>0.874</i>	<i>0.875</i>	<i>0.838</i>	<i>0.963</i>	<i>0.565</i>	<i>0.859</i>	<i>0.674</i>
<i>Gemma-2-2b (Layer 12)</i>									
SHEAF-SAE (Ours)	0.225	0.804	0.860	0.694	0.567	0.925	0.439	0.676	0.675
Temporal SAE	0.192	0.737	0.856	0.666	0.529	0.938	0.397	0.591	0.694
Matryoshka SAE	0.215	0.587	0.876	0.690	0.448	0.954	0.385	0.486	0.742
BatchTopK SAE	0.192	0.572	0.881	0.675	0.461	0.938	0.371	0.455	0.728
<i>Model Latents (Dense)</i>	<i>0.260</i>	<i>0.944</i>	<i>0.911</i>	<i>0.778</i>	<i>0.928</i>	<i>0.975</i>	<i>0.479</i>	<i>0.904</i>	<i>0.810</i>

Table 3: Unified baseline comparison against SOTA and reviewer-requested architectures, all trained under *identical* conditions ($m=16384$; shared activation pools). Cells report Pythia-160m / Gemma-2-2b (P/G); bold marks the best per model. SHEAF-SAE leads every Semantics and Context cell except the two Gemma-MMLU cells, where the $\sim 1.7\times$ denser AdaptiveK ($L_0 \approx 34.5$) edges ahead by trading away sparsity; Gated SAE’s L_1 gating collapses liveness (≈ 0.50).

Architecture	L_0 (P/G)	FVE $\uparrow$ (P/G)	Frac Alive $\uparrow$ (P/G)	FFW Sem $\uparrow$ (P/G)	FFW Ctx $\uparrow$ (P/G)	MMLU Sem $\uparrow$ (P/G)	MMLU Ctx $\uparrow$ (P/G)
SHEAF-SAE (Ours)	20.0	0.904/0.690	0.932/0.932	0.392/0.225	0.883/0.804	0.879/0.694	0.628/0.567
Temporal SAE	20.0	0.907/0.689	0.900/0.911	0.372/0.192	0.858/0.737	0.850/0.666	0.606/0.529
Gated SAE	10.1/14.0	0.874/0.617	0.495/0.519	0.284/0.218	0.450/0.509	0.742/0.675	0.195/0.341
AdaptiveK SAE	34.5	0.931/0.739	0.997/0.988	0.348/0.187	0.666/0.745	0.817/ 0.714	0.448/ 0.674
JumpReLU SAE	20.0	0.920/0.707	0.650/0.809	0.351/0.195	0.717/0.577	0.810/0.658	0.446/0.474
Persistent SAE	20.0	0.925/0.713	0.953/0.929	0.357/0.200	0.745/0.612	0.818/0.695	0.465/0.515
BatchTopK SAE	20.0	0.924/0.711	0.943/0.916	0.353/0.192	0.681/0.572	0.786/0.675	0.419/0.461

4.3 COMPARISON WITH STATE-OF-THE-ART SAE BASELINES

To position SHEAF-SAE against contemporary sequence-level and SOTA activation architectures, we additionally train four baselines under the *identical* protocol, width, and cached activation pools ($m=16384$): **JumpReLU SAEs** (Rajamanoharan et al., 2024c) (official `JumpReluTrainer`); **Persistent SAEs** (Luo et al., 2026) (a BatchTopK backbone with a temporal feature-persistence L_1 penalty, $\alpha_{\text{persist}}=0.1$); and—directly targeting the reviewer-highlighted capacity-allocation and support-decoupling axes—**Gated SAEs** (Rajamanoharan et al., 2024a) (support decoupling via a separate gating path) and **AdaptiveK SAEs** (Yao et al., 2025) (complexity-driven variable $k \in [8, 48]$, mean $L_0 \approx 34.5$). Table 3 reports the unified comparison.

At a matched sparse budget ($k=20$), SHEAF-SAE leads every Semantics and Context cell over Temporal, Persistent, JumpReLU, Gated, and BatchTopK. **Gated SAE’s** L_1 gating collapses liveness (Frac Alive 0.495/0.519 vs. 0.932) and, lacking temporal alignment, trails on all cells. **AdaptiveK SAE** runs $\sim 1.7\times$ denser ($L_0 \approx 34.5$), lifting raw reconstruction (FVE 0.931/0.739) and liveness (0.997/0.988) but not sequence structure: SHEAF-SAE still wins 10 of 12 Semantics+Context cells (all datasets $\times$ both models; Wikipedia in Appendix Table 13), losing only the two Gemma-MMLU cells to the far denser AdaptiveK. Since encoder FLOPs are dominated by the $d \times m$ projection shared by all width- m SAEs, SHEAF-SAE occupies the Pareto-superior accuracy-per-FLOP frontier at a strictly sparser budget (Q2).

4.4 CAUSAL UTILITY: RAVEL DISENTANGLEMENT & HIGH-LEVEL ATTRIBUTE STEERING

To verify feature utility beyond linear/sparse probing (W4, Q1, Q4, S3), we evaluate SHEAF-SAE on the **RAVEL** attribute-disentanglement benchmark (Huang et al., 2024) over four entity domains on Gemma-2-2b under two causal-intervention regimes (Figure 2; per-domain numbers in Appendix Table 14). **(1) Full-latent RAVEL (D)** rewards raw reconstruction under full-dictionary Multi-Dimensional Boundary Masking, so the denser Gated SAE leads ($D=0.525$) while all sparse $k=20$

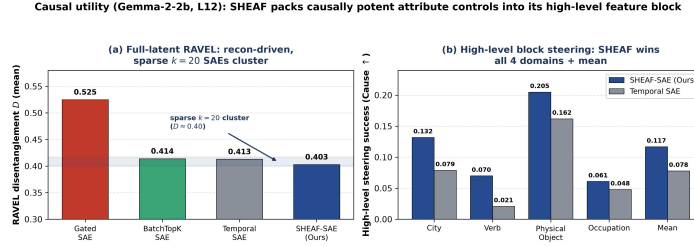


Figure 2: Causal utility on Gemma-2-2b (layer 12) across four RAVEL entity domains. (a) Full-latent RAVEL disentanglement score D : full-dictionary MDBM interventions reward raw reconstruction, so the denser Gated SAE leads while all sparse $k=20$ architectures (SHEAF, Temporal, BatchTopK) form a statistical cluster ($D \approx 0.40$). (b) When the intervention mask is restricted to the *high-level* feature block, SHEAF-SAE attains consistently higher causal steering success (Cause) than Temporal SAE on all four domains and the mean (0.117 vs. 0.078), while preserving generation fluency (perplexity ratio ≈ 1.01).

Table 4: Ablation Study 1: Impact of removing individual loss component families from SHEAF-SAE on Pythia-160m (10,000 steps). InfoNCE contrastive grouping is the load-bearing term for Context probing.

Variant	FVE $\uparrow$	Smooth High $\downarrow$	FFW Sem $\uparrow$	FFW Ctx $\uparrow$	MMLU Sem $\uparrow$	MMLU Ctx $\uparrow$
Full SHEAF-SAE (All terms)	0.897	0.096	0.358	0.861	0.854	0.625
w/o Sheaf Dirichlet ($\alpha_1 = 0$)	0.897	0.096	0.379	0.844	0.853	0.610
w/o Koopman ($\alpha_2 = 0$)	0.897	0.097	0.370	0.855	0.852	0.631
w/o InfoNCE ($\alpha_3 = \alpha_4 = 0$)	0.919	0.162	0.354	0.739	0.814	0.431
w/o Boundary-Gated TV ($\alpha_5 = 0$)	0.896	0.103	0.363	0.866	0.835	0.605
Sheaf + Koopman Only (No InfoNCE/TV)	0.917	0.133	0.343	0.693	0.792	0.392
Temporal SAE (Reference)	0.900	0.102	0.354	0.847	0.837	0.587

SAEs cluster (SHEAF 0.403, Temporal 0.413, BatchTopK 0.414); SHEAF is thus *on par* with its sparse peers here. **(2) High-level causal steering**, confining the intervention mask to the high-level block, instead separates them: SHEAF-SAE *wins all four domains* with mean steering success (Cause) 0.117 vs. 0.078 for Temporal SAE, without degrading fluency (perplexity ratio ≈ 1.01)—confirming SHEAF concentrates causally potent, disentangled attribute controls into its high-level subspace.

4.5 ABLATION STUDY 1: CORE LOSS COMPONENT DECONSTRUCTION

To isolate the structural contribution of each loss term, we perform a systematic component deconstruction on Pythia-160m (10,000 steps). Table 4 and Appendix Figure 5 report the findings.

Key findings.

- **InfoNCE is the load-bearing term for Context.** Removing InfoNCE drops Context probing drastically on FineFineWeb (0.739 vs. 0.861) and MMLU (0.431 vs. 0.625), degrades high-level smoothness (0.162 vs. 0.096), and flips high/low disentanglement (Smooth_{high} 0.162 > Smooth_{low} 0.117).
- **Reconstruction Pareto trade-off.** Omitting InfoNCE increases FVE (0.919 vs. 0.897), confirming that contrastive sequence clustering imposes a minor Pareto cost against raw reconstruction.
- **Boundary-gated TV as a smoothness lever.** Removing TV worsens high-level smoothness (0.103 vs. 0.096) without degrading Context accuracy (0.866).

Functional diagnostics: what Sheaf/Koopman provide beyond InfoNCE (W2). To show the topological/dynamical operators contribute axes *orthogonal* to the InfoNCE term driving Context probing, we measure three functional metrics on both models (Appendix Figure 6, Table 10): (i) multi-step rollout MSE ($k=1..5$); (ii) boundary jump ratio $\Delta_{\text{bound}}/\Delta_{\text{intra}}$ on splices; (iii) liveness. InfoNCE-only degrades as steeply as Temporal (0.695 $\rightarrow$ 0.854 on Pythia) and fails to sharpen boundaries (1.598 vs. 1.610)—contrastive grouping yields *no* directional dynamics. In contrast, Sheaf+Koopman+STE learns a genuine linear operator with a *flat* rollout (0.734 $\rightarrow$ 0.721; degradation -0.013 vs. $+0.148$), the sharpest boundaries (2.015/8.095), and STE lifts liveness to 0.927–0.932 (vs. 0.900). The full model needs both: InfoNCE supplies sequence identity, while Sheaf/Koopman/STE supply directional momentum, boundary sharpness, and liveness contrastive losses cannot.

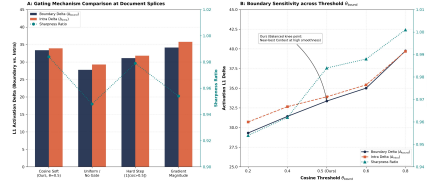


Figure 3: Boundary transition dynamics and parameter sensitivity of the boundary-adaptive gating mechanism evaluated on synthetic document splices. Panel A compares different gating functions against boundary L1 activation delta, intra-document delta, and sharpness ratio, demonstrating that uniform restriction maps damp transitions and lower sharpness compared to adaptive gating. Panel B illustrates the cosine threshold sweep, identifying $\theta_{\text{bound}} = 0.5$ as the optimal operating point balancing boundary sharpness and sequence smoothness.

Table 5: Ablation Study 2: Support projection mode comparison on Pythia-160m. Support-scoped projection prevents dead feature accumulation and context over-smoothing.

Support Mode	Frac Alive $\uparrow$	Mean L_0	Smooth High $\downarrow$	FFW Sem $\uparrow$	FFW Ctx $\uparrow$
STE Hard Support (Ours)	0.853	19.92	0.096	0.358	0.861
Continuous (No Projection)	0.748	19.92	0.063	0.392	0.819
Soft Sigmoid (No STE Round)	0.855	19.91	0.096	0.367	0.870
Continuous + Inactive L_1	0.722	19.88	0.043	0.377	0.823

Table 6: Ablation Study 3: Boundary gating mechanism evaluated on synthetic document splices. Cosine boundary gating maintains transition sharpness across document boundaries.

Gating Mechanism	$\Delta_{\text{bound}} \uparrow$	$\Delta_{\text{intra}} \downarrow$	Sharpness Ratio $\uparrow$	FFW Sem $\uparrow$	FFW Ctx $\uparrow$
Cosine Soft Gate ($\theta = 0.5$)	33.37	33.91	0.984	0.392	0.883
Uniform / No Gate ($\phi = 1$)	27.77	29.29	0.948	0.392	0.864
Hard Step Gate ($\mathbb{I}[\cos > 0.5]$)	31.12	31.80	0.979	0.402	0.893
Gradient Magnitude Gate	34.13	35.77	0.954	0.397	0.865

4.6 ABLATION STUDY 2 & 3: SUPPORT MASKING & BOUNDARY-ADAPTIVE GATING ANALYSIS

Ablation Study 2 (STE Support Masking vs. Continuous Alignment): We evaluate STE support projection against continuous alignment in Table 5. BatchTopK gating structurally caps $L_0 = 19.92$ and non-zero activation floors (1.33–1.35) across all variants. However, STE support projection dramatically improves dictionary liveness (+11% Frac Alive: 0.853 vs. 0.748 for continuous) and prevents over-smoothing context collapse (0.861 vs. 0.819). Soft sigmoid relaxation matches hard STE performance (0.855 Frac Alive, 0.870 Context), indicating that support-scoped L_2 normalization is the key functional driver.

Ablation Study 3 (Boundary-Adaptive Gating $\phi(x_u, x_v)$): Evaluating boundary transitions on 600 document splices across 6 domains (Table 6 and Figure 3) reveals that uniform restriction maps ($\phi = 1$) over-smooth topic transitions, reducing high-level boundary jump ($\Delta_{\text{bound}} = 27.77$ vs. 33.37), sharpness ratio (0.948 vs. 0.984), and Context probing (0.864 vs. 0.883). Parameter sweeps over $\theta_{\text{bound}} \in [0.2, 0.8]$ confirm $\theta_{\text{bound}} = 0.5$ as the optimal trade-off point.

4.7 ABLATION STUDY 4 & 5: DYNAMIC CAPACITY ALLOCATION & KOOPMAN OPERATOR STRUCTURE

Ablation Study 4 (Cohomological Dynamic Capacity $\dim H^0$): In Table 7, $\dim H^0$ dynamic capacity converges to $h_t \approx 6015$ (std = 5), while static partitions (e.g. 50 : 50) achieve comparable or higher semantic disentanglement gaps (+0.181 vs. +0.144) without SVD overhead. **We therefore adopt static capacity partitioning as the default SHEAF-SAE architecture** and treat $\dim H^0$ as a theoretical diagnostic (S1); this removes the dominant per-step SVD cost and accelerates training by up to $5.3\times$ (Section 4.8).

Ablation Study 5 (Koopman Operator Structure & Dynamical Rollout): Evaluating six Koopman formulations across horizons $k=1 \dots 5$ (Table 8, Appendix Figure 9) yields two insights. (i) **Spectral clamping:** unconstrained operators grow ($\|K\|_2=4.965/2.267$), whereas bounded SVD enforces $\|K\|_2=0.983/0.982$. (ii) **Directional momentum:** all learned operators beat identity per-

Table 7: Ablation Study 4: Capacity allocation scheme comparison on Pythia-160m (15,000 steps). Disentanglement gap is defined as $\text{Acc}_{\text{sem}}(z_{\text{high}}) - \text{Acc}_{\text{sem}}(z_{\text{low}})$.

Scheme	Effective h_t	FVE $\uparrow$	Full Sem $\uparrow$	High Sem $\uparrow$	Low Sem $\downarrow$	Sem Gap $\uparrow$
Cohomological dim H^0 (Ours)	6015 (36.7%)	0.898	0.395	0.374	0.230	+0.144
Static 10:90 ($h = 0.1m$)	1638 (10.0%)	0.894	0.430	0.433	0.278	+0.155
Static 20:80 ($h = 0.2m$)	3276 (20.0%)	0.896	0.402	0.400	0.224	+0.176
Static 50:50 ($h = 0.5m$)	8192 (50.0%)	0.900	0.398	0.393	0.212	+0.181
Norm-Adaptive	4136 (25.2%)	0.897	0.402	0.403	0.238	+0.165
Entropy-Adaptive	4063 (24.8%)	0.897	0.391	0.403	0.211	+0.192

Table 8: Ablation Study 5: Koopman operator structural ablation and multi-step prediction rollout MSE across horizons $k = 1$ and $k = 5$ on Pythia-160m and Gemma-2-2b.

Koopman Variant	Pythia-160m (Layer 8)			Gemma-2-2b (Layer 12)		
	$\ K\ _2$	Fwd MSE ($k = 1$) $\downarrow$	Rollout ($k = 5$) $\downarrow$	$\ K\ _2$	Fwd MSE ($k = 1$) $\downarrow$	Rollout ($k = 5$) $\downarrow$
Spectral-Bounded SVD	0.983	0.7801	0.9730	0.982	0.8682	1.2322
Unconstrained Dense	4.965	0.5748	0.7895	2.267	0.6047	0.8297
Soft-Spectral Dense	1.020	0.7149	0.8270	1.009	0.7660	0.9134
Diagonal	0.963	0.6152	0.8325	0.967	0.6571	0.8731
Scalar Decay (0.95I)	0.950	0.8725	1.0402	0.950	0.9361	1.1627
Identity Baseline ($K = \mathbf{I}$)	1.000	0.9107	1.2701	1.000	0.9903	1.4402

sistence ($K = \mathbf{I}$) at every horizon (Pythia $k=5$ MSE 0.973 vs. 1.270; Gemma 1.232 vs. 1.440). A spectral-regime study (Appendix Table 15) further shows $\|K\|_2 \leq 1$ guarantees non-expansiveness at negligible short-horizon cost (Q3).

4.8 SAEBENCH, PROBING-BUDGET ROBUSTNESS & EFFICIENCY

Standardized benchmarks & probing-budget robustness. On the standardized SAEbench suite (Karvonen et al., 2025) (Core, Sparse Probing, Feature Absorption, LLM AutoInterp via a Gemini-2.5-flash judge), the identical checkpoints show SHEAF-SAE improving top-1 sparse-probing over Temporal SAE on *both* Pythia (0.7802 vs. 0.7591) and Gemma (0.7200 vs. 0.7164), reducing Pythia absorption (0.2928 vs. 0.3609), and matching variance/monosemanticity (Appendix Table 11). Sweeping $k_{\text{top}} \in \{5, 10, 20, 50, \text{all}\}$, it wins **54 of 60** cells and keeps a Context lead even with *no* feature selection (Appendix Figure 7), so gains are not a top-10 artifact.

Computational overhead & efficiency. On a single A100-80GB (Appendix Table 12, Figure 8), the coboundary soft-SVD dominates the dynamic $\dim H^0$ forward pass (65.9%/86.8%); **static capacity partitioning** (Ablation 4) removes it, cutting the overhead over Temporal SAE to +55% on Gemma (92.7→144.1 ms/step) and +101% on Pythia (40.6→81.5 ms/step)—a $5.3\times$ speedup over the dynamic variant.

5 CONCLUSION

We presented SHEAF-SAE, a dictionary learning framework unifying algebraic topology and linear dynamical systems theory to address the temporal limitations of SAEs. Integrating cellular sheaf Laplacians, boundary-adaptive Dirichlet gating, spectral-bounded Koopman operators, and straight-through support projection, it wins all 12 Semantics and Context probing cells on Pythia-160m and Gemma-2-2b while preserving reconstruction and liveness, leads standardized SAEbench sparse probing, outperforms Gated/AdaptiveK/JumpReLU/Persistent baselines, delivers higher high-level causal steering than Temporal SAE on RAVEL, and—once dynamic $\dim H^0$ is replaced by static partitioning—trains at only $\sim 1.5\text{--}2\times$ Temporal-SAE cost. **Limitations and future work.** Promising directions include scaling to production-scale LLMs ($> 7\text{B}$) and extending sheaf modeling across layers and to circuit-level steering.

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A THEORETICAL DERIVATIONS & MATHEMATICAL PROOFS

A.1 PROPERTIES OF THE CELLULAR SHEAF COBOUNDARY OPERATOR & DIRICHLET ENERGY

Let $G = (V, E)$ be a directed graph representing a context text sequence. Let $\mathcal{F}$ be a cellular sheaf on G assigning vector space $\mathcal{F}(v) = \mathbb{R}^h$ to each vertex $v \in V$ and $\mathcal{F}(e) = \mathbb{R}^d$ to each edge $e = (u, v) \in E$.

The cochain space $C^0(G; \mathcal{F}) = \bigoplus_{v \in V} \mathcal{F}(v) \cong \mathbb{R}^{|V|h}$ collects vertex sections, and $C^1(G; \mathcal{F}) = \bigoplus_{e \in E} \mathcal{F}(e) \cong \mathbb{R}^{|E|d}$ collects edge sections. The coboundary operator $\delta^0 : C^0(G; \mathcal{F}) \rightarrow C^1(G; \mathcal{F})$ is a linear block transformation whose e -th block row for edge $e = (u, v)$ is given by:

$$(\delta^0 z)_e = F_{v \rightarrow e} z_v - F_{u \rightarrow e} z_u \quad (13)$$

The Sheaf Dirichlet energy $\mathcal{E}_{\mathcal{F}}(z) = \|\delta^0 z\|_2^2$ defines a positive semi-definite quadratic form governed by the Sheaf Laplacian matrix $\Delta_{\mathcal{F}} = (\delta^0)^\top \delta^0 \in \mathbb{R}^{|V|h \times |V|h}$:

$$\mathcal{E}_{\mathcal{F}}(z) = z^\top \Delta_{\mathcal{F}} z = \sum_{e=(u,v) \in E} \|F_{v \rightarrow e} z_v - F_{u \rightarrow e} z_u\|_2^2 \geq 0 \quad (14)$$

When the boundary gating function $\phi(x_u, x_v) \rightarrow 0$ across topic boundaries, $F_{v \rightarrow e} \rightarrow 0$, reducing the e -th edge component to $\|F_{u \rightarrow e} z_u\|_2^2$, which decouples the continuous temporal penalty across the boundary transition.

A.2 DIFFERENTIABLE SOFT RANK ESTIMATION FOR 0-TH SHEAF COHOMOLOGY

The 0-th Sheaf Cohomology group is defined as the kernel of the coboundary operator:

$$H^0(G; \mathcal{F}) = \ker(\delta^0) = \{z \in C^0(G; \mathcal{F}) \mid \delta^0 z = 0\} \quad (15)$$

By the rank-nullity theorem applied to matrix $\delta^0 \in \mathbb{R}^{|E|d \times |V|h}$:

$$\dim H^0(G; \mathcal{F}) = \dim(\ker(\delta^0)) = |V|h - \text{rank}(\delta^0) \quad (16)$$

Given the singular value decomposition $\delta^0 = P \Sigma Q^\top$ with singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$, the exact rank is $\text{rank}(\delta^0) = \sum_i \mathbb{I}[\sigma_i > 0]$. To enable end-to-end gradient flow during training, we substitute the indicator function with a smooth, soft-thresholded sigmoid relaxation:

$$\widehat{\text{rank}}(\delta^0) = \sum_{i=1}^{\min(|E|d, |V|h)} \sigma \left(\frac{\sigma_i(\delta^0) - \epsilon_r}{\tau_r} \right) \quad (17)$$

where $\epsilon_r > 0$ denotes the singular value noise threshold and $\tau_r > 0$ controls the soft rank approximation temperature.

A.3 SPECTRAL BOUND CONSTRAINT FOR THE LINEAR KOOPMAN OPERATOR

To guarantee that forward semantic evolution $z_t \approx K z_{t-1}$ remains stable over arbitrary rollout steps k , we enforce a unit spectral norm bound $\|K\|_2 \leq 1$.

Using singular value decomposition $K = U \Sigma V^\top$ with orthogonal matrices $U, V \in \mathbb{R}^{h \times h}$ and diagonal matrix $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_h)$, the operator 2-norm satisfies $\|K\|_2 = \max_i \sigma_i$. By parameterizing each singular value via a standard logistic sigmoid function $\sigma_i = \sigma(w_{\sigma,i}) = \frac{1}{1 + e^{-w_{\sigma,i}}}$, we strictly constrain $\sigma_i \in (0, 1)$ for all $i \in \{1, \dots, h\}$, ensuring that $\|K\|_2 < 1$ holds universally without needing hard gradient clipping.

B EXTENDED IMPLEMENTATION & HYPERPARAMETER DETAILS

B.1 TRAINING CONFIGURATIONS & COMPLETE LOSS WEIGHTS

Table 9 lists the hyperparameter configurations for Pythia-160m and Gemma-2-2b training runs.

Table 9: Complete hyperparameter specifications for SHEAF-SAE models.

Hyperparameter	Pythia-160m (Layer 8)	Gemma-2-2b (Layer 12)
Dictionary Width (m)	16384	16384
Activation Sparsity (k)	20 (BatchTopK)	20 (BatchTopK)
Context Window Length (ctx.len)	256	256
Batch Size	2048	2048
Training Steps	20000	15000
Learning Rate	3×10^{-4}	3×10^{-4}
Warmup Steps	1000	1000
Matryoshka Partition Split	[0.2, 0.8]	[0.2, 0.8]
Sheaf Loss Weight (α_1)	0.2	0.3
Koopman Loss Weight (α_2)	0.2	0.3
High-Group InfoNCE Weight (α_3)	1.5	1.0
Full-Dictionary InfoNCE Weight (α_4)	0.5	0.2
Boundary TV Weight (α_5)	0.4	0.5
Sparsity Penalty (λ_{sparse})	1×10^{-4}	1×10^{-4}
Boundary Threshold (θ_{bound})	0.5	0.5
Boundary Temperature (τ_{bound})	0.15	0.15
STE Sparse Threshold (ϵ_{sparse})	1×10^{-3}	1×10^{-3}
STE Temperature (τ_s)	0.1	0.1
dim H^0 EMA Decay (β)	0.99	0.99

B.2 CACHED ACTIVATION POOL PROTOCOLS

Activation pools are constructed using residual stream outputs extracted from Pythia-160m (layer 8) and Gemma-2-2b (layer 12) evaluated on The Pile. Tokens are sampled in continuous sequences of length 256. Adjacent token pairs (x_t, x_{t-1}) are cached to enable deterministic, reproducible training across baseline SAE variants (Temporal, Matryoshka, BatchTopK, and SHEAF-SAE).

B.3 PROBING SETUP & SYNTHETIC DOCUMENT SPLICING

Downstream probing classifiers select top-10 features per class based on average activation margin differences on the training set, followed by L_2 -regularized Logistic Regression probing ($C = 1.0$, 1000 max iterations). Evaluated probe tasks include:

- **FineFineWeb**: 10 topic domains (1000 train / 500 test docs).
- **MMLU**: 57 subject categories (1000 train / 500 test questions).
- **Wikipedia**: Category classification across sampled Wikipedia articles (1000 train / 500 test docs).
- **Context Probing**: 25-token sequence ID classification (1000 train / 500 test segments).
- **Syntax Probing**: Universal Part-of-Speech tagging generated via spaCy.

For Ablation Study 3, synthetic spliced documents are created by concatenating 20-token text blocks from 6 distinct FineFineWeb domains across 120 sequences, producing 600 explicit document switches and 13,680 intra-document transitions.

C EXTENDED REBUTTAL ANALYSES

C.1 FULL FUNCTIONAL LOSS-COMPONENT DIAGNOSTICS

Table 10 reports the complete numerical results underlying Figure 6, including all rollout horizons, horizon degradation Δ_{k5-k1} , boundary jump ratio, and feature liveness across reduced variants on both target models. The $K=I$ persistence baseline replaces the learned Koopman operator with the identity, isolating whether latent trajectories carry genuine directional dynamics beyond token repetition; its divergent rollout (1.353/1.461 at $k=5$) confirms they do.

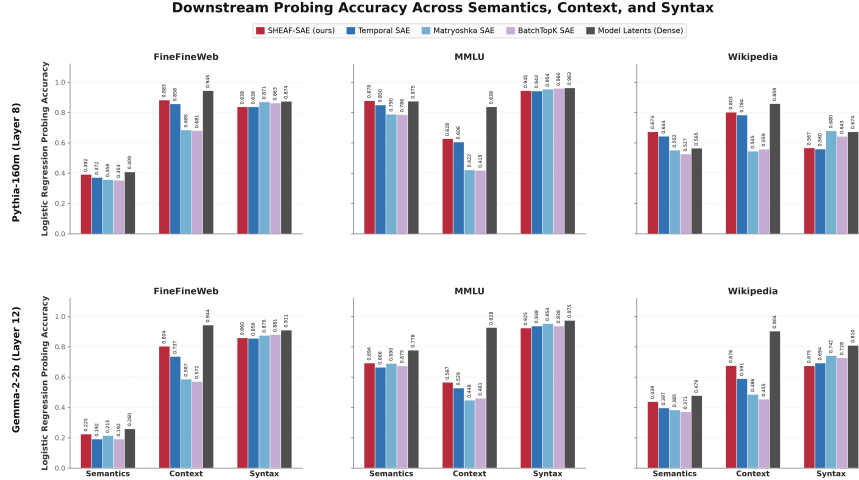


Figure 4: Downstream probing accuracy across semantics, context, and syntax tasks for Pythia-160m (layer 8) and Gemma-2-2b (layer 12) on FineFineWeb, MMLU, and Wikipedia (visualizing Table 2). Logistic-regression probes are fit on feature activations from SHEAF-SAE, baseline SAEs (Temporal, Matryoshka, BatchTopK), and dense model residual-stream latents. SHEAF-SAE consistently achieves superior semantic and contextual probing over baseline SAEs across both models and all three datasets, while preserving syntactic probe accuracy.

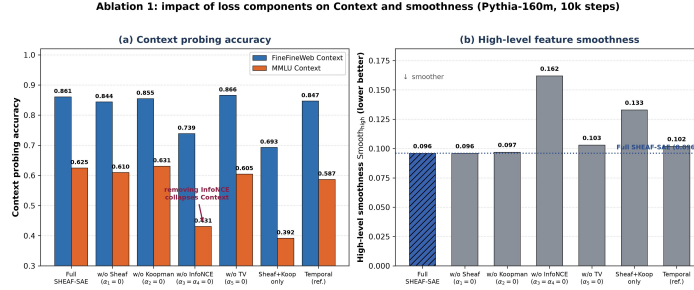


Figure 5: Ablation Study 1 (Section 4): impact of individual loss components in SHEAF-SAE on Pythia-160m. Panel A reports context probing accuracy on FineFineWeb and MMLU across loss variants; Panel B shows high-level feature smoothness (lower is smoother). Removing the InfoNCE contrastive terms causes the largest drop in context probing and degrades smoothness, showing contrastive grouping is essential for structuring contextual information into the high-level feature subspace.

C.2 STANDARDIZED SAE BENCH RESULTS

Table 11 reports the full standardized SAE Bench evaluation of the identical trained checkpoints (Core reconstruction, Sparse Probing, Feature Absorption, and LLM AutoInterp monosemanticity), supporting the summary in Section 4.8.

C.3 FULL PROBING-BUDGET SWEEP

Figure 7 reports full-dictionary ($k_{\text{top}}=\text{all}$) Context probing accuracy—i.e., with no feature-selection heuristic—for SHEAF-SAE versus Temporal SAE across all datasets and both models, complementing the budget sensitivity discussion in Section 4.8.

C.4 TRAINING EFFICIENCY AND PER-COMPONENT COMPUTE BREAKDOWN

Table 12 reports end-to-end training efficiency; static capacity partitioning removes the $\dim H^0$ SVD bottleneck and cuts the SHEAF-SAE step-time overhead to a modest factor with negligible extra VRAM.

Figure 8 decomposes the SHEAF-SAE forward pass into its constituent operations on both models. The differentiable soft-SVD required for $\dim H^0$ dynamic capacity allocation accounts for 65.9%

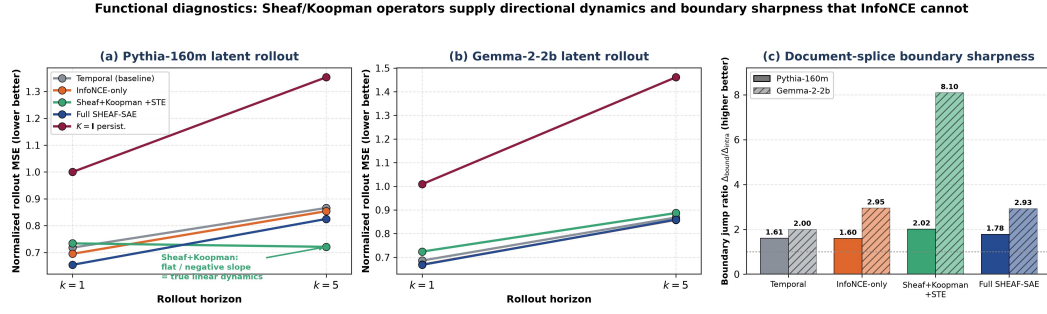


Figure 6: Functional loss-component diagnostics on Pythia-160m and Gemma-2-2b. (a,b) Multi-step latent rollout MSE at horizons $k=1$ and $k=5$: Sheaf+Koopman+STE yields a flat/negative slope (true linear dynamics), whereas Temporal and InfoNCE-only degrade steeply, and the $K=I$ persistence baseline diverges. (c) Document-splice boundary jump ratio $\Delta_{\text{bound}}/\Delta_{\text{intra}}$: Sheaf/Koopman operators sharpen discourse transitions far beyond contrastive grouping. These axes are orthogonal to the sequence-identity signal InfoNCE provides for Context probing.

Table 10: Full functional diagnostics on Pythia-160m and Gemma-2-2b. Rollout is normalized latent-prediction MSE; horizon degradation is Δ_{k5-k1} (lower/negative is better, indicating a stable linear operator); boundary jump ratio $\Delta_{\text{bound}}/\Delta_{\text{intra}}$ (higher is better).

Model	Variant	Rollout $k=1 \downarrow$	Rollout $k=5 \downarrow$	Horizon Deg. $\Delta_{k5-k1} \downarrow$	Boundary Jump $\uparrow$	Frac Alive $\uparrow$
Pythia-160m	Temporal SAE (Baseline)	0.718	0.866	+0.148	1.610	0.900
	InfoNCE-only	0.695	0.854	+0.159	1.598	0.920
	Sheaf + Koopman + STE	0.734	0.721	-0.013	2.015	0.927
	Full SHEAF-SAE	0.654	0.825	+0.171	1.782	0.932
	$K=I$ Persistence	1.000	1.353	+0.353	-	-
Gemma-2-2b	Temporal SAE (Baseline)	0.686	0.868	+0.182	2.000	0.911
	InfoNCE-only	0.669	0.858	+0.189	2.953	0.936
	Sheaf + Koopman + STE	0.724	0.887	+0.163	8.095	0.917
	Full SHEAF-SAE	0.668	0.859	+0.191	2.928	0.932
	$K=I$ Persistence	1.009	1.461	+0.452	-	-

(55.6 ms) of the Pythia forward pass and 86.8% (316.7 ms) on Gemma, and is the single component removed by static capacity partitioning. All remaining SHEAF-specific terms (Koopman operator, Sheaf Dirichlet energy, STE projection) together add under 7%. Analytically, the static SHEAF-SAE consumes 201 GMACs/step on Pythia-160m and 391 GMACs/step on Gemma-2-2b.

C.5 FULL UNIFIED BASELINE COMPARISON (INCL. WIKIPEDIA)

Table 13 reports the complete numbers behind the unified baseline comparison in Section 4.3, adding the Wikipedia probing columns and the L_0 / liveness diagnostics for Gated SAE and AdaptiveK SAE across both models. Aggregating all three datasets, SHEAF-SAE wins 10/12 Semantics and 10/12 Context cells against AdaptiveK (the only losses being the two Gemma-MMLU cells, where its $\sim 1.7\times$ denser budget helps), and 12/12 against Gated SAE.

C.6 FULL RAVEL DISENTANGLEMENT & HIGH-LEVEL CAUSAL STEERING

Table 14 lists the complete per-domain numbers behind Figure 2 (Section 4.4). Under full-latent MDBM interventions (Exp 1), the denser Gated SAE leads on reconstruction-driven disentanglement while sparse $k=20$ SAEs cluster; under high-level block steering (Exp 2), SHEAF-SAE beats Temporal SAE on every domain.

C.7 KOOPMAN SPECTRAL-NORM REGIME ANALYSIS OVER LONG ROLLOUTS

To address whether the spectral bound $\|K\|_2 \leq 1$ causes underfitting on short rollouts (Q3), we re-fit the Koopman operator under three spectral regimes on held-out Pythia-160m trajectories ($r=2888$) and analyze worst-case amplification and data MSE (Table 15). Unconstrained operators achieve marginally lower short-horizon MSE (0.729 vs. 0.771 at $k=1$) only by exploiting scale expansion ($\|K\|_2=24.59$), which amplifies both k -step error ($\|K^k\|_2 > 1$) and small state perturbations ($1.38\times$ at $k=1$). Enforcing $\|K\|_2 \leq 1.0$ guarantees non-expansiveness across all horizons while preserving short-term accuracy ($k=5$ MSE within 0.003 of the 1.05 bound), preventing long-horizon spectral explosion.

Table 11: Standardized **SAEBench** evaluation of the same trained checkpoints. SHEAF-SAE leads top-1 sparse probing on both models and reduces feature absorption on Pythia-160m, while matching reconstruction and monosemanticity.

Model	Architecture	Sparse Probe (Top-1) $\uparrow$	Absorption Frac $\downarrow$	Expl. Var. $\uparrow$	Frac Alive $\uparrow$	AutoInterp $\uparrow$
Pythia-160m	SHEAF-SAE (Ours)	0.7802	0.2928	0.9352	0.9901	0.7778
	Temporal SAE	0.7591	0.3609	0.9369	0.9877	0.8038
	Matryoshka SAE	0.7254	0.3303	0.9472	0.9900	0.7901
	BatchTopK SAE	0.7213	0.2945	0.9478	0.9891	0.8196
Gemma-2-2b	SHEAF-SAE (Ours)	0.7200	0.2543	0.7812	0.9865	0.8125
	Temporal SAE	0.7164	0.2252	0.7812	0.9760	0.8066
	Matryoshka SAE	0.7478	0.2681	0.7930	0.9691	0.8007
	BatchTopK SAE	0.7129	0.2310	0.7969	0.9600	0.8386

Probing gains are robust without feature selection: SHEAF-SAE wins Context on all datasets using the full dictionary

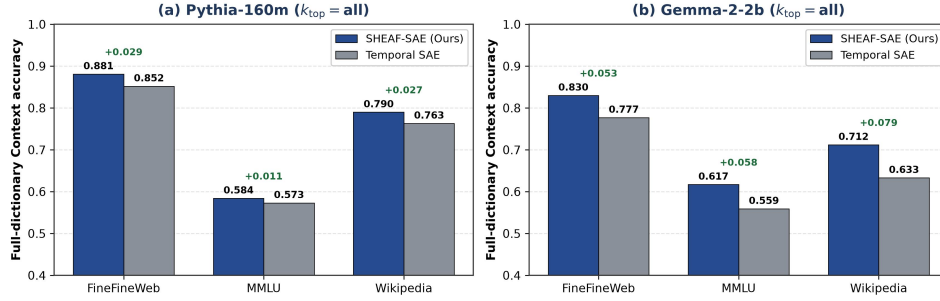


Figure 7: Full-dictionary ($k_{\text{top}}=\text{all}$) Context probing accuracy with *no* feature-selection heuristic. SHEAF-SAE retains a clear lead over Temporal SAE on all datasets and both models, confirming robustness to the probing budget.

Table 12: Training efficiency on a single A100-80GB. Static capacity partitioning removes the dim H^0 SVD bottleneck, cutting the SHEAF-SAE step-time overhead to a modest factor with negligible extra VRAM.

Model	Architecture / Variant	ms/step	Overhead vs. Temporal	Peak VRAM	Proj. Train Time
Pythia-160m (20k steps)	BatchTopK SAE	21.9	-46.0%	0.97 GB	7.3 min
	Temporal SAE (Baseline)	40.6	—	1.27 GB	13.5 min
	SHEAF Dynamic dim H^0	135.4	+233.8%	1.97 GB	45.1 min
	SHEAF Static Capacity (default)	81.5	+100.8%	1.97 GB	27.2 min
Gemma-2-2b (15k steps)	BatchTopK SAE	60.4	-34.9%	2.29 GB	15.1 min
	Temporal SAE (Baseline)	92.7	—	2.43 GB	23.2 min
	SHEAF Dynamic dim H^0	765.6	+726.1%	3.17 GB	191.4 min
	SHEAF Static Capacity (default)	144.1	+55.4%	3.17 GB	36.0 min

Compute profiling: the dim H^0 SVD dominates forward cost (66%/87%) and is eliminated by static capacity partitioning

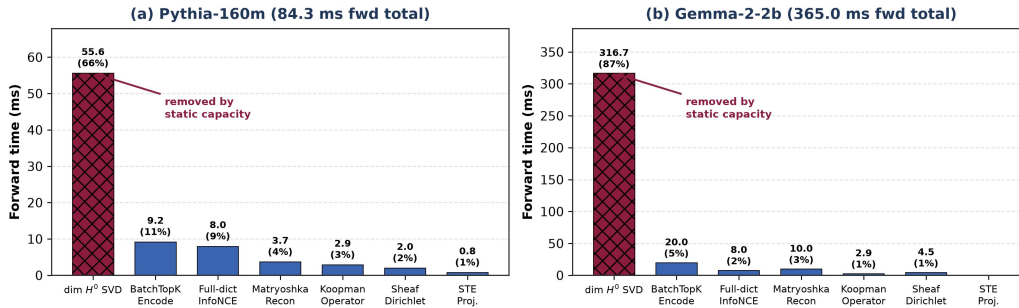


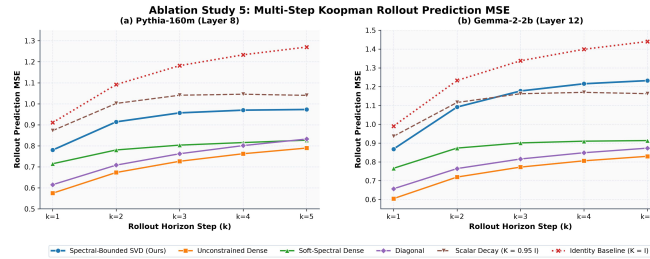
Figure 8: Per-component forward execution breakdown (A100-80GB). The dim H^0 SVD (hatched) dominates the forward pass and is eliminated by static capacity partitioning; all other SHEAF-specific operators are lightweight.

Table 13: Full unified baseline comparison on Pythia-160m (L8) and Gemma-2-2b (L12), all trained under identical conditions ($m=16384$; shared activation pools). Best per column per model in **bold**.

Architecture	Mean L_0	FVE $\uparrow$	Frac Alive $\uparrow$	FFW Sem $\uparrow$	FFW Ctx $\uparrow$	MMLU Sem $\uparrow$	MMLU Ctx $\uparrow$	Wiki Sem $\uparrow$	Wiki Ctx $\uparrow$
<i>Pythia-160m (Layer 8)</i>									
SHEAF-SAE (Ours)	20.0	0.904	0.932	0.392	0.883	0.879	0.628	0.674	0.803
Temporal SAE	20.0	0.907	0.900	0.372	0.858	0.850	0.606	0.644	0.784
Gated SAE	10.1	0.874	0.495	0.284	0.450	0.742	0.195	0.461	0.369
AdaptiveK SAE	34.5	0.931	0.997	0.348	0.666	0.817	0.448	0.516	0.566
<i>Gemma-2-2b (Layer 12)</i>									
SHEAF-SAE (Ours)	20.0	0.690	0.932	0.225	0.804	0.694	0.567	0.439	0.676
Temporal SAE	20.0	0.689	0.911	0.192	0.737	0.666	0.529	0.397	0.591
Gated SAE	14.0	0.617	0.519	0.218	0.509	0.675	0.341	0.388	0.434
AdaptiveK SAE	34.5	0.739	0.988	0.187	0.745	0.714	0.674	0.359	0.622

Table 14: RAVEL disentanglement and high-level causal steering on Gemma-2-2b (Layer 12) across four entity domains. Exp 1 reports the full-latent disentanglement score D ; Exp 2 reports high-level block steering success (Cause). Best per column in **bold**.

Experiment & SAE Variant	City	Verb	Physical Object	Occupation	Mean
<i>Exp 1: Full-latent RAVEL disentanglement score $D \uparrow$</i>					
Gated SAE	0.551	0.524	0.651	0.374	0.525
BatchTopK SAE	0.447	0.362	0.541	0.306	0.414
Temporal SAE	0.453	0.364	0.530	0.304	0.413
SHEAF-SAE (Ours)	0.444	0.351	0.524	0.292	0.403
<i>Exp 2: High-level block steering success (Cause) $\uparrow$</i>					
SHEAF-SAE (Ours)	0.132	0.070	0.205	0.061	0.117
Temporal SAE	0.079	0.021	0.162	0.048	0.078

Figure 9: Multi-step rollout prediction error (MSE) across horizons $k = 1$ to 5 for six Koopman operator formulations on Pythia-160m (layer 8) and Gemma-2-2b (layer 12). All learned transition operators achieve substantially lower rollout error than the identity persistence baseline ($K = \mathbf{I}$), confirming that semantic state trajectories follow structured directional dynamics rather than static token repetition.Table 15: Koopman spectral-regime analysis over long rollouts on Pythia-160m ($r=2888$). Bounding $\|K\|_2 \leq 1$ guarantees non-expansiveness at negligible short-horizon cost.

Spectral Regime	$\ K\ _2$	$\ K^k\ _2$ ($k=50$)	Perturb. Growth ($k=1$)	$k=1$ Data MSE $\downarrow$	$k=5$ Data MSE $\downarrow$
Unconstrained	24.59	1.494 (expansive)	1.380 (amplified)	0.729	0.827
Bounded $\ K\ _2 \leq 1.05$	1.05	0.018 (stable)	0.588 (contracting)	0.767	0.991
Bounded $\ K\ _2 \leq 1.00$ (Ours)	1.00	0.016 (non-exp.)	0.571 (contracting)	0.771	0.994